

65/130

33.3-1:

$$T = 37.8^\circ\text{C}$$

$$P = 101.3 \text{ kPa}$$

$$p_A = 3.59 \text{ kPa}$$

15/15

$$(a) H = \frac{18.02(3.59)}{28.97(101.3 - 3.59)} = 0.0229 \text{ kg H}_2\text{O/kg air}$$

(b) From steam tables,

$$p_s = 5.6236 + (37.8 - 35) \left[\frac{7.384 - 5.6236}{40 - 35} \right] = 6.609 \text{ kPa}$$

$$H_s = \frac{18.02(6.61)}{28.97(101.3 - 6.61)} = 0.0434 \text{ kg H}_2\text{O/kg air}$$

$$H_p = \frac{0.0229}{0.0434} \times 100 = 52.8\%$$

$$(c) H_R = \frac{3.59}{6.61} \times 100 = 54.3\%$$

33.1-3: Dew Pt = 15.6°C (60°F) $\Rightarrow$ 100% saturation
Dry Bulb Temp. = 65.6°C (150°F)

15/15

$H = 0.0125 \text{ kg H}_2\text{O/kg dry air}$ (from the humidity chart)

$\rightarrow$ Actual humidity of air @ 65.6°C

Thus, $H_p = 5\%$ (from humidity chart)

$$C_s = 1.005 + 1.88(0.0125) = 1.0285 \text{ kJ/kg dry air} \cdot \text{K}$$

(or)

$$= 1.0285 \times 10^3 \text{ J/kg} \cdot \text{K}$$

$$C_s = 0.24 + 0.45(0.0125) = 0.246 \text{ Btu/lbm dry air} \cdot ^\circ\text{F}$$

$$v_H = \frac{(2.83 \times 10^{-3}) + (4.56 \times 10^{-3})(0.0125)}{0.978 \text{ m}^3/\text{kg dry air}}$$

(or)

$$v_H = \frac{0.0252 + (0.0405)(0.0125)}{15.681 \text{ ft}^3/\text{lbm dry air}}$$

3.3-9:

Dry bulb temp. = 37.8°C

Wet bulb temp. = 26.7°C

0/
20

33.6-1:

$$t_1 = \frac{L_s}{AR} (X_1 - X_2)$$

$$h = 50.8 = 0.0508 \text{ m}$$

$$R = 2.05$$

$$X_1 = 0.55$$

$$X_2 = 0.30$$

$$\frac{L_s}{A} \rightarrow \frac{L_s}{2A(0.0508)}$$

$$= \frac{24.4}{2(0.0508)} = 240.16$$

$$t = \frac{240.16}{2.05} (0.55 - 0.30)$$

$$= 29.288 \text{ h}$$

10
/ 20

3.6-3:

dry bulb temp = $40^\circ\text{C} = T_b$

$x_1 = 0.35 \text{ kg H}_2\text{O} / \text{kg dry solid}$

Wet bulb temp = $29.4^\circ\text{C} = T_w$

$x_2 = 0.22 \text{ kg H}_2\text{O} / \text{kg dry solid}$

$A = 0.61 \text{ m} \times 0.61 \text{ m} = 0.3721 \text{ m}^2$

$b = 25.4 \text{ mm} = 0.0254 \text{ m}$

$v = 3.05 \text{ m/s}$

$L_s = 11.34 \text{ kg dry solid}$

(a)

$H = 0.014 \text{ kg H}_2\text{O} / \text{kg dry solid}$

$$V_H = [(2.83 \times 10^{-3}) + (4.56 \times 10^{-3})(0.014)](40 + 273)$$

$$= 0.9636 \text{ m}^3 / \text{kg dry air}$$

$$m = 1.0 \text{ kg dry air} + 0.014 \text{ kg H}_2\text{O} = 1.014$$

$$\rho = \frac{1.014}{0.9636} = 1.0523 \text{ kg/m}^3$$

$$G = (3.05)(3600)(1.0523) = 11554 \text{ kg/h} \cdot \text{m}^2$$

$$h = 0.0204 (11554)^{0.68} = 0.0431 \text{ W/m}^2 \cdot \text{K}$$

$h_w = 2400$ from steam table

$$R_c = \frac{0.0431}{2400} (40 - 29.4)(3600) = 1.978 \text{ kg/h} \cdot \text{m}^2$$

$$t = \frac{11.34}{(0.3721)(1.978)} (0.35 - 0.22) = 2.003 \text{ hrs}$$

$$(b) \frac{t_1}{t_2} = \frac{b_1}{b_2} \Rightarrow t_2 = \frac{t_1 b_2}{b_1} = \frac{(2.003)(44.5 \text{ mm})}{25.4 \text{ mm}}$$

$$t_2 = 3.509 \text{ hr}$$

33.8-2:

$$T = 60^\circ\text{C}$$

$$V = 3.05 \text{ m/s}$$

$$Z_s = 25.4 \text{ mm} = 0.0254 \text{ m}$$

$$Z_m = 1.59 \text{ mm} = 0.0159 \text{ m}$$

$$k_m = 43.3 \text{ W/m}\cdot\text{K}$$

$$k_s = 1.125 \text{ W/m}\cdot\text{K}$$

$$H = 0.010 \text{ kg H}_2\text{O/kg dry solid}$$

$$\epsilon = 0.94$$

$$T_s = 104.4^\circ\text{C}$$

$$\rho = \frac{1 \text{ kg dry air} + 0.010 \text{ kg H}_2\text{O}}{0.974} = 1.037 \text{ kg/m}^3$$

$$G = (3.05)(3600)(1.037) = 11380 \text{ kg/h}\cdot\text{m}^2$$

$$h = 0.0204 (11380)^{0.8} = 35.87 \text{ W/m}^2\cdot\text{K}$$

$$\lambda_s =$$

33.10-3:

$$V = 0.914 \text{ m/s}$$

$$T = 82.2^\circ\text{C}$$

$$H = 0.01 \text{ kg H}_2\text{O/kg dry air}$$

$$\rho_s = 1522 \text{ kg/m}^3$$

$$\varepsilon = 0.35$$

$$X_1 = 0.90 \text{ kg H}_2\text{O/kg solid}$$

$$X_2 = 0.45 \text{ kg H}_2\text{O/kg solid}$$

$$X^* < X_2$$

